

Ankur Desai

(248) 657 – 3805 | ardusa05@gmail.com | ardusa.github.io | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- GPA: 3.5 / 4.0
- **Involvement:** Vice President & Ambassador of Claude Builder Club, Eagle Scout
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Michigan State University Unmanned Systems

Software Engineer

Oct. 2025 – Present

East Lansing, MI

- Engineered autonomous drone payload delivery system on Nvidia Jetson, integrating MAVLink and Orange+ flight controller, capable of releasing packages from 80+ feet altitude with a precision of ± 3 feet.
- Built machine learning pipeline: cleaned and labeled 5K aerial images, trained and tuned YOLOv11 model, and validated real-time inference on embedded CUDA hardware with TensorRT through regression tests.
- Benchmarked Python inference against a refactored C++ implementation, achieving a 36% optimization in detection latency on embedded platforms.

MindMosaic — Relationship Powered Note Taking Tool

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Directed a team of 4 to build a full-stack note-taking app with a chapter-and-page model, layered Markdown editing, and visual concept mapping via annotated links.
- Developed a React web app (Vite, Tailwind CSS, marked) with OAuth2 authentication and established CI/CD pipelines, achieving 94% code coverage.
- Deployed a dual database system (PostgreSQL + Neo4j), using Neo4j for rapid breadth-first search lookups; improved query speed by 48% and reduced server load by 30%.

PROJECTS

Mira — Distributed Intelligence Tool | *FastAPI, Next.js, AWS (EC2, RDS)* May 2025 – Nov. 2025

End-to-end voice intelligence system, aggregating real-time audio streams to execute context-aware automation

- Architected a thread-safe multi-tenant service registry using LRU eviction and TTL-based resource cleanup to manage lifecycle for 100+ concurrent networks, preventing memory leaks in long-running server instances.
- Developed scalable NER and sentiment analysis pipelines (spaCy, Hugging Face Transformers) in Dockerized FastAPI services, orchestrated CI/CD to AWS EC2/RDS.
- Improved Whisper model accuracy from 91% to 98% through spectral gating, denoising, and gain control.
- Built a production hybrid search pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim), implementing 5-minute conversation boundary detection to optimize context retrieval accuracy.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy*

Feb. 2025

Head-to-head combat game utilizing pose estimation and gesture recognition **Winner:** Computer Vision Track

- Engineered a low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Developed a custom vector-based physics engine handling AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash

AI & Computer Vision PyTorch, TensorFlow, MediaPipe, Pyannote.audio, Sentence-Transformers

Full-Stack Development React, Next.js, FastAPI, Electron, SQLAlchemy, Node.js

DevOps & Cloud AWS (Lambda, EC2, RDS), Docker, GitHub Actions, Linux, PostgreSQL